

The parkinsonian neurotoxin, MPP⁺ induces phosphorylated c-Jun in dopaminergic neurons of mesencephalic cultures

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Abstract

The stress-activated protein kinase (SAPK) cascade serves a critical role in the apoptotic death of neuronal cells in response to a variety of cellular stresses. Recent *in vitro* and *in vivo* evidence has directly implicated this kinase in the death of dopaminergic nigral neurons in the MPTP model of Parkinson's disease (PD). To assess the involvement of c-Jun, a key transcription factor substrate of SAPK (also known as c-Jun N-terminal kinase, or JNK) in the MPTP-induced death of dopaminergic nigral neurons, we determined the ability of MPP⁺, the active toxin metabolite of MPTP, to induce the phosphorylated form of c-Jun in dopaminergic neurons in nigral (ventral mesencephalon) cultures. At a dose of MPP⁺ that specifically induces apoptotic changes in nuclear morphology in tyrosine hydroxylase-positive (dopaminergic) cells in these cultures, MPP⁺ induces nuclear phospho-c-Jun immunoreactivity (IR). The peak induction of phospho-c-Jun IR was observed 16 h after beginning MPP⁺ exposure, and preceded the maximal induction of apoptotic nuclear changes by approximately 8 h. These data support an important role for the SAPK/JNK pathway including its c-Jun transcriptional target in the apoptotic death of dopaminergic nigral neurons in the MPTP model of PD. © 2001 Elsevier Science Ltd. All rights reserved.

Keywords: Stress-activated protein kinase; c-Jun N-terminal kinase; Apoptosis; 1-methyl-4-phenylpyridinium; Ventral mesencephalon

1. Introduction

Although a role for apoptosis in Parkinson's disease (PD) has not been proven, it has been implicated by the repeated observation of apoptotic features in the death of dopaminergic nigral neurons in the 1-methyl-4-phenyl-1,2,3,6-tetrahydropyridine (MPTP) model of PD and in PD itself (reviewed in [1]). Evidence that new gene expression is required for apoptosis [2] suggests that signaling pathways linking noxious extracellular signals to gene transcription may play a pivotal role in apoptotic neuronal death. Stress-activated protein kinase (SAPK; also known as c-Jun N-terminal kinase, or JNK) and its transcription factor targets are prominent candidate mediators of apoptotic cell death in response to a wide variety of cellular stressors in neuronal [3–5] as well as non-neuronal cells [6], although SAPK is not required for some forms of apoptotic cell death [7]. The recent demonstration that CEP-1347, a relatively specific pharmacological inhibitor of JNK activation, can markedly inhibit the MPTP-induced loss of dopaminergic mesencephalic neurons in rodents [8]

and in primates [9] has prompted clinical trials of this agent in PD with the goal of slowing progression of the disease.

The leucine zipper transcription factor c-Jun is phosphorylated (at serine residues 63 and 73) and consequently activated by SAPK/JNK, and its activated form can be a critical mediator of apoptotic cell death in the nervous system (reviewed in [10]). However other candidate mediators of apoptotic cell death, including phosphorylated forms of transcription factors such as ATF-2 and p53, and of cytoplasmic proteins such as tau and bcl-2 may also be induced by SAPK/JNK [10,11]. Thus, despite strong evidence for SAPK/JNK activation mediating apoptotic cell death in the MPTP model of PD, c-Jun remains only one of several SAPK/JNK targets that may carry on the neuronal cell death signaling triggered by this stress kinase pathway. Here we consider further the role of c-Jun activation by determining the effect of MPP⁺, the active toxin metabolite of MPTP, on nuclear phosphorylated c-Jun immunoreactivity (P-Jun IR) in dopaminergic neurons in culture.

2. Materials and methods

Primary cultures from rat E-17 ventral mesencephalon

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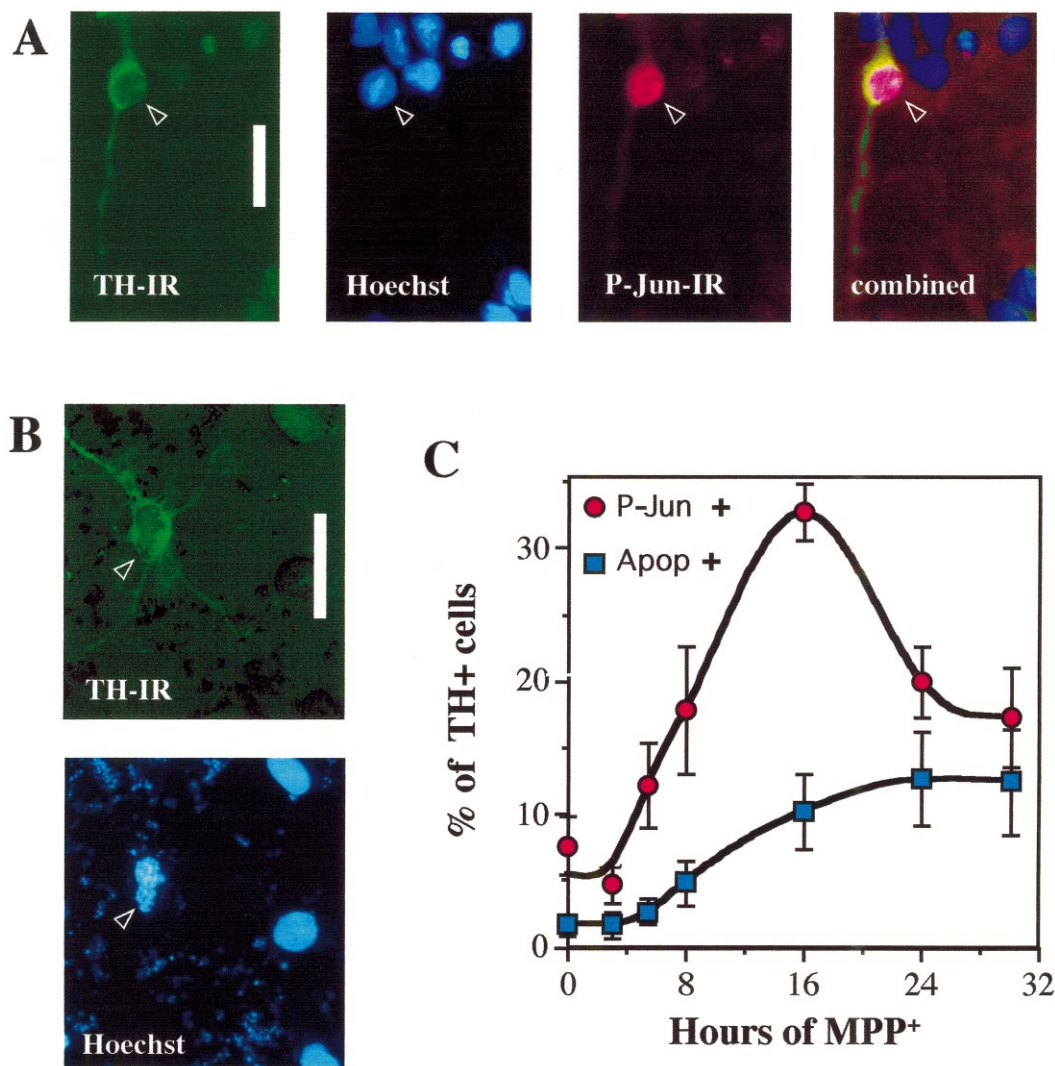


Fig. 1. MPP^+ sequentially induces P-Jun IR and apoptotic morphological changes in nuclei of TH + neurons in ventral mesencephalon cultures. (A) Induced P-Jun IR in TH + cells was restricted to the nucleus as shown in a representative field of an E-17 rat ventral mesencephalon culture exposed to MPP^+ for 16 h. The fluorescence photomicrographs of the same dopaminergic neuron (arrowheads) were taken through three filter sets optimized for Hoechst 33258 and for the respective chromophores coupled to TH and P-Jun IR. A combination image was generated by superimposing these photomicrographs. (B) Apoptotic nuclear (chromatin) morphology was scored as condensed fragmented Hoechst 33258 staining as shown in a representative dopaminergic neuron (arrowhead) in a ventral mesencephalon culture exposed to MPP^+ for 24 h. A photomicrograph of TH IR was digitally filtered for Hoechst 33258 staining (which was also detected by the filter set used for TH IR) to generate the image shown. The length of the white bars shown in (A) and (B) represents 25 μ . (C) Time courses for MPP^+ -induced P-Jun IR, and MPP^+ -induced apoptotic features of nuclear Hoechst 33258 staining in TH + neurons. Compared to control values (at 0 h), $P < 0.05$ for all P-Jun + and Apop + levels at and beyond 8 h; $n \sim 5$.

(which gives rise to the substantia nigra in adults) were seeded on polyethylenimine-coated eight-well Permanox chamber slides (Lab-tek) and maintained for five days in serum-free medium (F-12/DMEM with B-27 supplements). Cultures were then treated with or without MPP^+ for various times up to 30 h. After fixation with 4% paraformaldehyde in phosphate-buffered saline, double immunocytochemistry for tyrosine hydroxylase (TH) with a mouse monoclonal anti-TH (1:500; Boehringer-Mannheim), and then for phospho(63 ser)-c-Jun with a rabbit polyclonal antibody (1:1000; New England Biolabs) was performed. After treatment with each primary antibody, it was in turn labeled with a Cy2- or Cy3-coupled anti-mouse or anti-rabbit

secondary antibody (Jackson Immunoresearch), respectively. Lastly, chromatin was stained with Hoechst 33258 (5 μ g/ml for 5 min) before final rinses and coverslip mounting over 1:1 glycerol:phosphate buffered saline. Under fluorescence microscopy wells were systematically analyzed in a blinded manner for the percentage of TH-positive neurons that also display elevated P-Jun IR or a pattern of nuclear chromatin staining consistent with cell death by apoptosis [12,13]. Nuclei were scored as positive for apoptotic features of Hoechst 33258 staining if the normally faint homogeneous staining pattern was replaced by condensed bright nuclear fragments [14]. The 'Principles of laboratory animal care' (NIH publication no. 86-23,

revised 1985) were followed in accordance with our institutional policies.

3. Results and discussion

At a concentration of 10 μM , MPP^+ gradually induced P-Jun IR in cultured TH-positive (dopaminergic) neurons with a significant increase apparent by 8 h and a peak induction (at least four-fold over basal levels) noted after the first 16 h, before a partial decrease over the next 14 h [Fig. 1(A) and (C)]. Induced P-Jun IR was restricted to the nucleus [Fig. 1(A)]. Under these conditions, the induction by MPP^+ was specific for dopaminergic neurons, presumably due to their expression of the dopamine transporter which is required for MPP^+ toxicity [15]. At 10 (and 100) μM MPP^+ , induction of P-Jun IR was detected only in TH IR + cells, whereas at a higher MPP^+ concentration (1000 μM), TH – as well as TH + mesencephalic cells displayed a marked induction after 24 h of treatment (three-fold and 10-fold over baseline values in TH – and TH + cells, respectively) (data not shown).

After an initial lag period, MPP^+ treatment (10 μM) also significantly induced apoptotic changes in nuclear morphology (condensation and fragmentation of Hoechst 33258-stained chromatin) in TH + cells in culture [Fig. 1(B)]. Maximal induction of these apoptotic features was noted 24 h after exposing ventral mesencephalon cultures to MPP^+ , and thus follows the peak in phospho-Jun IR induction [Fig. 1(C)]. Of note, in one experiment P-Jun IR/TH + cells were also scored in a blinded manner for the presence of apoptotic nuclear morphology. None were observed before 16 h of MPP^+ exposure; but after 16, 24 and 30 h of MPP^+ treatment 3.6, 5.4 and 8.0% of P-Jun IR/TH + cells, respectively, also displayed apoptotic nuclear features. Although the rarity of these apoptotic P-Jun IR/TH + cells (two out of 55 cells, two out of 37, and three out of 37 P-Jun IR/TH + cells, respectively) precluded statistical analysis, it nevertheless suggests the transience of nuclear P-Jun prior to the apoptotic death of dopaminergic neurons. These data however do not exclude the possibility that P-Jun IR is expressed in dopaminergic cells that are largely distinct from those undergoing apoptosis.

Thus we have demonstrated MPP^+ -induced nuclear P-Jun IR in cultured dopaminergic neurons in advance of apoptotic changes. Our findings strengthen the evidence that the SAPK/JNK pathway mediates dopaminergic toxicity in the MPTP model of PD. Furthermore, they support a role for c-Jun as a critical substrate of SAPK/JNK in MPP^+ -induced dopaminergic neuron death. Activation of c-Jun by N-terminal phosphorylation enhances the transcription of multiple genes whose increased expression contributes to apoptotic cell death. The best-characterized target of phospho-Jun is *c-jun* itself [16]. Through this auto-regulatory mechanism activation of the SAPK/JNK cascade leads to increased expression of *c-jun*, which is closely associated with neuronal cell death

[10]. Of note, a delayed increase in total c-Jun (independent of its phosphorylation state) has been demonstrated in dopaminergic neurons in vivo after systemic MPTP treatment [17], consistent with increased *c-jun* expression. Thus, our data strengthen the evidence that c-Jun as well as upstream elements in the SAPK/JNK cascade represent important targets of new therapeutic strategies for PD.

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